agricultural biodiversity from the twin threats of genetic engineering and patents.

## Agro-biodiversity

Agricultural biodiversity (also known as agro-biodiversity) has resulted from the selective breeding, over thousands of years, of plants and animals taken from the wild. These practices led to the extraordinary genetic diversity of different breeds of crops and domesticated animals. For example, a grass in the genus *Oryza* was first cultivated for rice in Asia between 8,200 and 13,500 years ago; today, there are more than 40,000 varieties of this rice in existence. Intrinsic to agro-biodiversity are the many non-harvested species that support production. These include microorganisms in the soil, species that feed on pests, and pollinators. Through the ages, the skills and knowledge of millions of farmers have shaped this biodiversity.

From the late 1960s, a technology transfer to the developing world included high-yield varieties of cereals in association with chemical fertilizers, pesticides, and herbicides, mechanization, and more efficient irrigation. Known as the “Green Revolution,” this transformation shifted the focus of agriculture in the developing world away from biodiversity to higher crop yields. New Green Revolution crops such as “miracle rice” (IR8) boosted production, but there was a downside. As more emphasis was placed on fewer productive strains, the genetic base of traditional seed varieties for grains, potatoes, fruits, vegetables, and cotton declined.

The United Nations Food and Agriculture Organization estimates that 75 percent of crop biodiversity has been lost from the world’s fields. Some environmentalists have argued that traditional varieties are more compatible with local farming conditions, cheaper for farmers to use, and more environmentally sustainable than new, high-yield varieties. Additionally, many of the new strains are patented by the companies that created them.